



UPPSALA
UNIVERSITET

Some more details

Implementation of functions



Functions with *one* argument

Input : `exp(2)`

Result: 7.38905609893065

Input : `exp(log(1+1)=y)`

Result: 2.0

Input : `3*exp(y) - cos(2*PI)`

Result: 5.0

Input : `fib(6)`

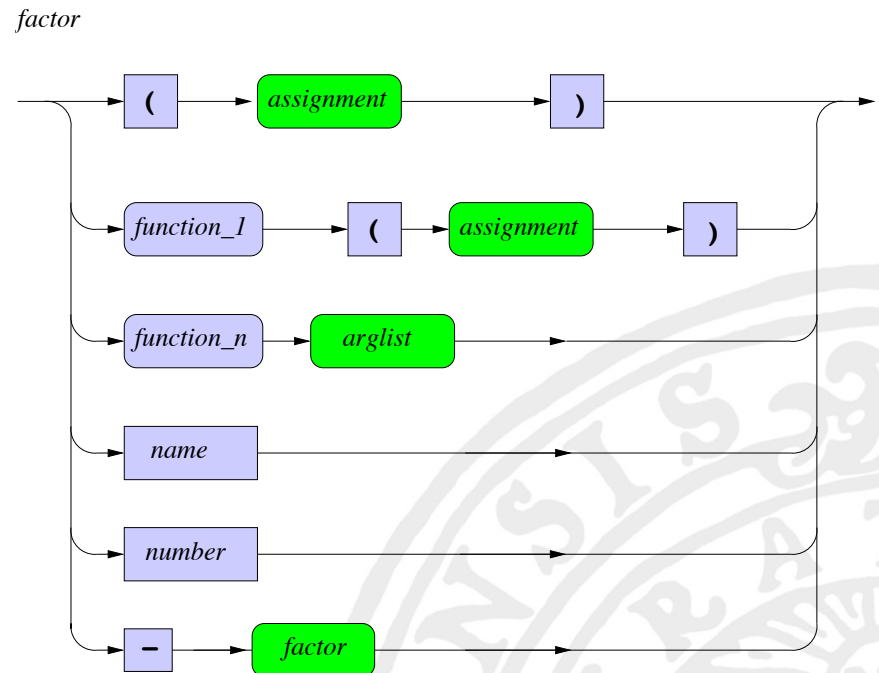
Result: 8

Input : `fac(2+3)`

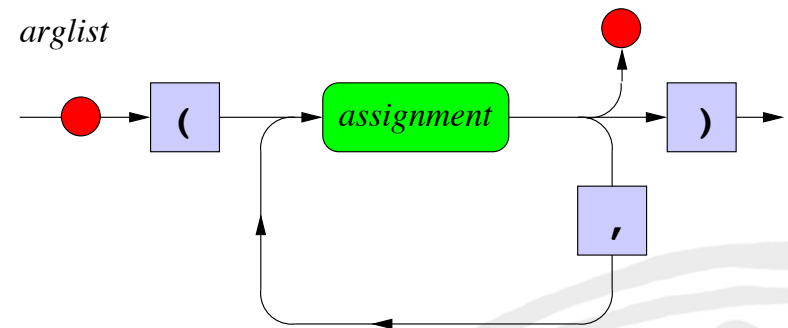
Result: 120

Input : `sin 2`

*** Syntax error: Expected '(' after function name
Error occurred at '2' just after 'sin'



Functions with *several* arguments



Input : `max(exp(0.8), 2*log(1.5))`

Result: 2.225540928492468

Input : `mean(1,2,3,4,5)`

Result: 3.0

Input : `min()`

*** Syntax error: Expected number, word or '('

Error occurred at ')' just after '('

Input : `max 2,3,4`

*** Syntax error: Expected '(' after function name

Error occurred at '2' just after 'max'

Evaluation errors in functions

- Illegal *value* of the argument. Example: `log(2*10-30)`.
- Illegal *type* of the argument. Exempel: `fib(3.5)`.

Such things must be checked by the function itself. Don't do that in the parser!

You can of course use Python's functions, but it can be good to write your own `log` which checks the argument and then uses `math.log`.



UPPSALA
UNIVERSITET

The end

